

Qualification Pack



Lab Chemist (Rubber)

QP Code: RSC/Q2301

Version: 4.0

NSQF Level: 4

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket
District Centre
New Delhi - 110017 || email: vishaka.v@rcpsdc.in

Qualification Pack

Contents

RSC/Q2301: Lab Chemist (Rubber)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
RSC/N2302: Collect rubber sample and prepare equipment	5
RSC/N2303: Conduct testing of rubber products at various stages of production.	14
RSC/N2304: Analysis, Reporting and Recording of Test Results.	23
RSC/N5007: Follow health & safety protocol	31
DGT/VSQ/N0102: Employability Skills (60 Hours)	39
Assessment Guidelines and Weightage	46
<i>Assessment Guidelines</i>	46
<i>Assessment Weightage</i>	47
Acronyms	48
Glossary	49

Qualification Pack

RSC/Q2301: Lab Chemist (Rubber)

Brief Job Description

A Lab Chemist - Rubber conducts chemical and physical testing of raw materials and rubber products to ensure quality as per standards. He/She also compiles SQC data like Cp and CpK to support process control and maintain product uniformity.

Personal Attributes

This role requires a self-driven individual with integrity, strong technical and interpersonal skills, and the ability to learn quickly. The person should interpret test findings effectively and take appropriate corrective actions.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N2302: Collect rubber sample and prepare equipment](#)
2. [RSC/N2303: Conduct testing of rubber products at various stages of production.](#)
3. [RSC/N2304: Analysis, Reporting and Recording of Test Results.](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Non - Tyre
Occupation	Testing and Quality Control
Country	India
NSQF Level	4
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/4322.0201

Qualification Pack

Minimum Educational Qualification & Experience	12th Class (or Equivalent) OR 10th grade pass with 2 year NTC plus 1 year NAC/CITS (or Equivalent) OR 10th Class with 3 Years of experience Relevant OR Previous relevant Qualification of NSQF Level with 3 Years of experience Relevant
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	4.0
Reference code on NQR	QG-04-RI-04032-2025-V2-RSDC
NQR Version	2

Qualification Pack

RSC/N2302: Collect rubber sample and prepare equipment

Description

This unit is about collecting samples and preparing equipments to carry out lab testing of raw-materials, chemicals, compounds, semi-finished and finished rubber products.

Elements and Performance Criteria

Sample collection

To be competent, the user/individual on the job must be able to:

- PC1.** draw sample of the material from the lot to be tested as per standard procedures (sop).
- PC2.** sampling to be done as per process flow sheet with control points.
- PC3.** identify the sample by labeling/numbering as per sop and recording in the log book /pc and assigning a sequential local lab number (category wise).
- PC4.** inspect the sample for visual defects including it has to free from any fm etc and approve the sample dimensionally.
- PC5.** ensure that sample is suitable for test/analysis
- PC6.** process sample should be collected after the process is established not from beginning of the process start
- PC7.** take / collect sample as per sop only. sample may be in form of raw material process material or ready product used in processing.

Sample integrity

To be competent, the user/individual on the job must be able to:

- PC8.** identify the defect/problem in inappropriate sample
- PC9.** identify factors that affect the integrity of sample (eg. temperature, moisture, contamination, contact with liquids
- PC10.** ensure samples are well protected and are free from picking up any moisture, foreign matter etc.
- PC11.** maintain integrity of the sample as per sop.
- PC12.** identify and store sample for future/further testing as per sop.

Equipment readiness

To be competent, the user/individual on the job must be able to:

- PC13.** identify the most appropriate equipment for testing as per the sop.
- PC14.** set up appropriate equipment/apparatus to be used for testing correctly as per sop / is / iso / international standard.
- PC15.** calibrate /verify/validate the testing equipment periodically as per sop and see monitoring sticker is available in respective equipment.
- PC16.** identify defective equipment/apparatus and steps to be taken as per sop
- PC17.** keep hand tools like gloves, knife , emery paper, sample cutter etc ready before starting the tests.

Housekeeping and Safety

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC18.** ensure health & safety in laboratory
- PC19.** ensure availability of safety accessories including eye wash station.
- PC20.** ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.
- PC21.** handle the equipment properly
- PC22.** ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.
- PC23.** use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations.
- PC24.** escalate matters in case of any accidents, spills etc.
- PC25.** comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys quality policies and acceptance standards for raw materials, processed and final product.
- KU2.** organisational/ national and international coding system of raw material, compounds and products.
- KU3.** chemicals used in the industry and their function including msds
- KU4.** different quality management systems
- KU5.** principles of good laboratory practices applicable in the workplace
- KU6.** material disposal procedure, importance of appropriate disposal of material and implications of not following the material disposal procedure.
- KU7.** quality and damage checks to be done and importance of the same.
- KU8.** importance of identifying non-conforming products and storage of the same.
- KU9.** risk and impact of not following defined procedures/work instructions .
- KU10.** escalation matrix for reporting identified issues.
- KU11.** types of documentation in organization and importance of the same.
- KU12.** records to be maintained and implications of non-maintenance of the same.
- KU13.** company manual and from where to attain it.
- KU14.** importance of housekeeping & good shop floor practices (e.g.3s/5s).
- KU15.** health, safety and environment guidelines, legislation and regulations as applicable.
- KU16.** personal protection(which protective equipment to be used and how).
- KU17.** impact of poor practices on health, safety and environment.
- KU18.** potential hazards and actions to minimize the same.
- KU19.** escalation matrix and escalation procedure for reporting hazards.
- KU20.** the usage of different fire extinguisher.
- KU21.** impact of various practices on cost, quality, productivity, delivery and safety.
- KU22.** handover/ takeover the equipment/ work area as per companys sop
- KU23.** knowledge of basic chemistry and simple mathematical calculations

Qualification Pack

- KU24.** knowledge on different standard reference from national and international material.
- KU25.** role of different raw materials in rubber compounding, processing/ product manufacturing and performance.
- KU26.** understanding of role of the testing equipments including setting.
- KU27.** testing equipments and related test methods and purpose of tests.
- KU28.** calibration requirements for test equipment.
- KU29.** procedures for storing samples.
- KU30.** specifications of materials tested and its importance in the release system.
- KU31.** national/international standard test methods for different materials.
- KU32.** preparation of standard chemical reagents for testing.
- KU33.** standard method of drawing samples and preparing them for testing.
- KU34.** how to assess whether a sample is suitable for testing.
- KU35.** methods/techniques used for labeling samples.
- KU36.** procedure (sop) to be followed in case the sample is unfit for test.
- KU37.** the methods that can be used for controlling test variables.
- KU38.** implications (impact on internal/external customers) of defective products, materials or components.
- KU39.** the material safety data sheets (msds) for all the materials used for the experiments that one is conducting.
- KU40.** procedures for storing and retention period for samples.
- KU41.** factors that adversely affect integrity of the sample.
- KU42.** use of computer/application software.
- KU43.** units of measurement.
- KU44.** response to emergencies e.g. power failures, fire and system failures and manual intervention to avoid disaster.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and communicate details of work done to appropriate people using written/typed report or computer based record/electronic mail
- GS2.** maintain proper records as per given format.
- GS3.** read and understand manuals, health and safety instructions, memos, reports, job cards etc.
- GS4.** read images, graphs, diagrams.
- GS5.** understand the various coding systems as per company norms.
- GS6.** communication with upstream and downstream teams.
- GS7.** communicate with job owners like sample originating section, supplier etc.
- GS8.** work in a team and other behavioral skills required to support the small group activities (eg. quality circle, cross functional team, suggestion scheme).
- GS9.** disclose information only to those who have the right and need to know it.

Qualification Pack

- GS10.** communicate confidential and sensitive information discretely to authorized person as per sop.
- GS11.** practice honesty with respect to company property and time.
- GS12.** communicate with people in a form and manner and using language that is open and respectful.
- GS13.** resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust.
- GS14.** take responsibility for completing ones own work assignment.
- GS15.** take initiative to enhance/learn skills in oness area of work
- GS16.** the capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse ones learning.
- GS17.** is open to new ways of doing things.
- GS18.** the capacity to envisage and articulate personal goals
- GS19.** avoid absenteeism.
- GS20.** act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations.
- GS21.** be punctual
- GS22.** take a decision for any change/issue based on earlier successes (documented previous history) on similar issues
- GS23.** work out changes in case a new improved machine/equipment is added in the process or any new material /chemical is developed replacing existing one.
- GS24.** make changes in cycle time due to improved process.
- GS25.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management.
- GS26.** consult the peer group and superiors to arrive at a favourable decision.
- GS27.** use of standard available problem solving techniques for decision making
- GS28.** review and analyze the process steps to check on system non adherence and non conformity.
- GS29.** review the current sop and other standards for continuous improvement to facilitate decision making.
- GS30.** take a calculated risk with minimum losses
- GS31.** organize samples and records properly.
- GS32.** communicate results as per organizational procedure.
- GS33.** perform analysis in given time line.
- GS34.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required).
- GS35.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS36.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS37.** communicate effectively to the superior/customer for any delay in supplies tothe clients.
- GS38.** work towards fulfilling the customers requirement as per their demand.
- GS39.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level

Qualification Pack

- GS40.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS41.** maintain good/cordial relation with customers.
- GS42.** work on the feedback received from customer regarding the product.
- GS43.** interpret quality for tested samples
- GS44.** apply problem-solving approaches in different situations
- GS45.** application of statistics
- GS46.** use of computer/application software
- GS47.** proper collection of waste material.
- GS48.** identify defects in the material and communicate it at the earliest and suggest improvements(if any) in process/material based on experience.
- GS49.** identify any issues affecting the material, equipment or surroundings.
- GS50.** escalate issues that cannot be solved as per the troubleshooting/company manual
- GS51.** apply appropriate technique/method for various types of products to meet desired purpose.
- GS52.** interpret data and analyse results
- GS53.** suggest improvements(if any) in process/product/materials based on experience.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sample collection</i>	13	14	-	-
PC1. draw sample of the material from the lot to be tested as per standard procedures (sop).	4	-	-	-
PC2. sampling to be done as per process flow sheet with control points.	-	2	-	-
PC3. identify the sample by labeling/numbering as per sop and recording in the log book /pc and assigning a sequential local lab number (category wise).	3	2	-	-
PC4. inspect the sample for visual defects including it has to free from any fm etc and approve the sample dimensionally.	3	3	-	-
PC5. ensure that sample is suitable for test/analysis	-	3	-	-
PC6. process sample should be collected after the process is established not from beginning of the process start	-	2	-	-
PC7. take / collect sample as per sop only. sample may be in form of raw material process material or ready product used in processing.	3	2	-	-
<i>Sample integrity</i>	12	12	-	-
PC8. identify the defect/problem in inappropriate sample	3	3	-	-
PC9. identify factors that affect the integrity of sample (eg. temperature, moisture, contamination, contact with liquids	3	3	-	-
PC10. ensure samples are well protected and are free from picking up any moisture, foreign matter etc.	3	2	-	-
PC11. maintain integrity of the sample as per sop.	-	2	-	-
PC12. identify and store sample for future/further testing as per sop.	3	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Equipment readiness</i>	17	14	-	-
PC13. identify the most appropriate equipment for testing as per the sop.	3	2	-	-
PC14. set up appropriate equipment/apparatus to be used for testing correctly as per sop / is / iso / international standard.	4	3	-	-
PC15. calibrate /verify/validate the testing equipment periodically as per sop and see monitoring sticker is available in respective equipment.	4	3	-	-
PC16. identify defective equipment/apparatus and steps to be taken as per sop	3	3	-	-
PC17. keep hand tools like gloves, knife , emery paper, sample cutter etc ready before starting the tests.	3	3	-	-
<i>Housekeeping and Safety</i>	8	10	-	-
PC18. ensure health & safety in laboratory	-	1	-	-
PC19. ensure availability of safety accessories including eye wash station.	-	1	-	-
PC20. ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.	-	2	-	-
PC21. handle the equipment properly	1	1	-	-
PC22. ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.	2	2	-	-
PC23. use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations.	1	1	-	-
PC24. escalate matters in case of any accidents, spills etc.	2	1	-	-
PC25. comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).	2	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	50	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2302
NOS Name	Collect rubber sample and prepare equipment
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	Lab Chemist
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/10/2026
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N2303: Conduct testing of rubber products at various stages of production.

Description

This unit is about carrying out lab testing of raw-materials, chemicals, compounds, semi-finished and finished rubber products.

Elements and Performance Criteria

Sample testing

To be competent, the user/individual on the job must be able to:

- PC1.** ensure that the reagents and materials used for testing are of standard quality and procured from approved source.
- PC2.** identify appropriate tests or follow guidelines given
- PC3.** carry out tests as per sop.
- PC4.** ensure that test methods conform to the required quality and accuracy of testing.

Testing for raw material

To be competent, the user/individual on the job must be able to:

- PC5.** carry out testing for raw material as per requirement.
- PC6.** specify gravity, volatile loss, moisture, particle size, surface area, assay, ash content, melting point, boiling point, softening point, ph as per type of raw material including specific tests (as mentioned in rm specification) and/or plant specification.
- PC7.** carry out volumetric, gravimetric or instrumental tests as per type of rm
- PC8.** visual examination of rm including tests with respect to appearance /color and odour.

Testing for batch release

To be competent, the user/individual on the job must be able to:

- PC9.** carry out testing for batch release as per the requirement.
- PC10.** carry out testing as per specific gravity, dispersion, cure characteristic properties, rheometry, mooney viscosity, mooney scorch, hardness, tensile, elongation at break, modulus, tear strength, rebound resilience, as per sop.

Testing for cured compound

To be competent, the user/individual on the job must be able to:

- PC11.** carry out testing for cured compound as per requirement: specific gravity, dispersion, cure characteristic properties, rheometry, mooney viscosity, hardness, tensile, elongation, modulus, tear, low temperature properties, ozone resistance, aging properties, abrasion resistance, rebound resilience, set properties, electrical properties, tear resistance, dispersion, heat build up, swelling as per requirement /standards.

Testing for finished products

To be competent, the user/individual on the job must be able to:

- PC12.** carry out testing for finished products as per requirement/guidelines provided.

Qualification Pack

- PC13.** carry out testing as per specific gravity, stress-strain properties (ts, m300, eb, tear) ,abrasion resistance (volume loss),set properties (compression, tension, permanent etc.),fatigue test (crack initiation, crack growth),rebound resilience, hardness testing (shore a/shore d),hysteresis properties.
- PC14.** carry out testing as per swelling properties, aging (hot air, humid, salt, oxygen bomb),low temperature properties, adhesion (fabric-rubber, rubberrubber, rubber-metal), electrical properties, dimension and checks endurance, uniformity, x-ray analysis, in accordance is / iso / international standards and sop.

Housekeeping and Safety

To be competent, the user/individual on the job must be able to:

- PC15.** ensure health & safety in laboratory
- PC16.** ensure availability of safety accessories including eye wash station.
- PC17.** ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.
- PC18.** handle the equipment properly
- PC19.** ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.
- PC20.** conduct the experiments wearing the appropriate attire such as safety goggles, gloves, closed toe shoes, long pants, tied hair.
- PC21.** use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations
- PC22.** escalate matters in case of any accidents, spills etc.
- PC23.** comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys quality policies and acceptance standards for raw materials, processed and final product.
- KU2.** organisational/ national/ international coding system of raw material, compounds and products.
- KU3.** chemicals used in the industry and their function.
- KU4.** different quality management systems.
- KU5.** principles of good laboratory practices applicable in the workplace.
- KU6.** material disposal procedure, importance of appropriate disposal of material and implications of not following the material disposal procedure
- KU7.** quality and damage checks to be done and importance of the same
- KU8.** importance of identifying non-conforming products and storage of the same.
- KU9.** risk and impact of not following defined procedures/work instructions.
- KU10.** escalation matrix for reporting identified issues.
- KU11.** types of documentation in organization and importance of the same.
- KU12.** records to be maintained and implications of non-maintenance of the same.
- KU13.** company manual and from where to attain it.

Qualification Pack

- KU14.** importance of housekeeping & good shop floor practices (e.g.3s/5s).
- KU15.** health, safety and environment guidelines, legislation and regulations as applicable
- KU16.** personal protection(which protective equipment to be used and how).
- KU17.** impact of poor practices on health, safety and environment.
- KU18.** potential hazards and actions to minimize the same
- KU19.** escalation matrix and escalation procedure for reporting hazards.
- KU20.** the usage of different fire extinguisher.
- KU21.** impact of various practices on cost, quality, productivity, delivery and safety.
- KU22.** handover/ takeover the equipment/ work area as per companys sop.
- KU23.** knowledge of basic chemistry and simple mathematical calculations
- KU24.** knowledge on different standard reference material.
- KU25.** role of different raw materials in rubber compounding, processing/ product manufacturing and performance.
- KU26.** understanding of role of the testing equipments.
- KU27.** testing equipments and related test methods and purpose of tests.
- KU28.** calibration requirements for test equipment.
- KU29.** procedures for storing samples.
- KU30.** specifications of materials tested and its importance in the release system.
- KU31.** national/international standard test methods for different materials.
- KU32.** preparation of standard chemical reagents for testing.
- KU33.** standard method of drawing samples and preparing them for testing.
- KU34.** how to assess whether a sample is suitable for testing .
- KU35.** methods/techniques used for labeling samples.
- KU36.** procedure (sop) to be followed in case the sample is unfit for test.
- KU37.** the methods that can be used for controlling test variables.
- KU38.** implications (impact on internal/external customers) of defective products, materials or components.
- KU39.** the material safety data sheets (msds) for all the materials used for the experiments that one is conducting.
- KU40.** procedures for storing and retention period for samples.
- KU41.** factors that adversely affect integrity of the sample.
- KU42.** statistical analysis of test data.
- KU43.** how to obtain and interpret records, charts, specifications, equipment manuals, history/technical support reports and other documents.
- KU44.** methods and techniques involved in evaluating information.
- KU45.** use of computer/application software.
- KU46.** units of measuremen.
- KU47.** Response to emergencies e.g. Power failures, fire and system failures and manual intervention to avoid disaster.

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** record and communicate details of work done to appropriate people using written/typed report or computer based record/electronic mail
- GS2.** maintain proper records as per given format.
- GS3.** read and understand manuals, health and safety instructions, memos, reports, job cards etc.
- GS4.** read images, graphs, diagrams.
- GS5.** understand the various coding systems as per company norms.
- GS6.** communication with upstream and downstream teams.
- GS7.** communicate with job owners like sample originating section, supplier etc.
- GS8.** disclose information only to those who have the right and need to know it.
- GS9.** communicate confidential and sensitive information discretely to authorized person as per sop.
- GS10.** practice honesty with respect to company property and time.
- GS11.** communicate with people in a form and manner and using language that is open and respectful.
- GS12.** resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust.
- GS13.** take responsibility for completing ones own work assignment.
- GS14.** take initiative to enhance/learn skills in oness area of work
- GS15.** the capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse ones learning.
- GS16.** is open to new ways of doing things
- GS17.** the capacity to envisage and articulate personal goals
- GS18.** avoid absenteeism.
- GS19.** act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations.
- GS20.** be punctual.
- GS21.** take a decision for any change/issue based on earlier successes (documented previous history) on similar issues.
- GS22.** work out changes in case a new improved machine/equipment is added in the process or any new material /chemical is developed replacing existing one.
- GS23.** make changes in cycle time due to improved process.
- GS24.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management.
- GS25.** consult the peer group and superiors to arrive at a favourable decision.
- GS26.** use of standard available problem solving techniques for decision making .
- GS27.** review and analyze the process steps to check on system non adherence and non conformity.
- GS28.** review the current sop and other standards for continuous improvement to facilitate decision making.
- GS29.** take a calculated risk with minimum losses.
- GS30.** organize samples and records properly

Qualification Pack

- GS31.** communicate results as per organizational procedure.
- GS32.** perform analysis in given time line.
- GS33.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS34.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS35.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS36.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS37.** work towards fulfilling the customers requirement as per their demand.
- GS38.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS39.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS40.** maintain good/cordial relation with customers.
- GS41.** work on the feedback received from customer regarding the product.
- GS42.** interpret quality for tested samples.
- GS43.** apply problem-solving approaches in different situations.
- GS44.** application of statistics.
- GS45.** use of computer/application software.
- GS46.** proper collection of waste material.
- GS47.** identify defects in the material and communicate it at the earliest and suggest improvements(if any) in process/material based on experience.
- GS48.** identify any issues affecting the material, equipment or surroundings.
- GS49.** escalate issues that cannot be solved as per the troubleshooting/company manual
- GS50.** apply appropriate technique/method for various types of products to meet desired purpose.
- GS51.** interpret data and analyse results.
- GS52.** suggest improvements(if any) in process/product/materials based on experience.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sample testing</i>	14	-	-	-
PC1. ensure that the reagents and materials used for testing are of standard quality and procured from approved source.	4	-	-	-
PC2. identify appropriate tests or follow guidelines given	3	-	-	-
PC3. carry out tests as per sop.	4	-	-	-
PC4. ensure that test methods conform to the required quality and accuracy of testing.	3	-	-	-
<i>Testing for raw material</i>	18	7	-	-
PC5. carry out testing for raw material as per requirement.	5	2	-	-
PC6. specify gravity, volatile loss, moisture, particle size, surface area, assay, ash content, melting point, boiling point, softening point, ph as per type of raw material including specific tests (as mentioned in rm specification)and/or plant specification.	4	2	-	-
PC7. carry out volumetric, gravimetric or instrumental tests as per type of rm	4	-	-	-
PC8. visual examination of rm including tests with respect to appearance /color and odour.	5	3	-	-
<i>Testing for batch release</i>	8	2	-	-
PC9. carry out testing for batch release as per the requirement.	4	1	-	-
PC10. carry out testing as per specific gravity, dispersion, cure characteristic properties, rheometry, mooney viscosity, mooney scorch, hardness, tensile, elongation at break, modulus, tear strength, rebound resilience, as per sop.	4	1	-	-
<i>Testing for cured compound</i>	6	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. carry out testing for cured compound as per requirement: specific gravity, dispersion, cure characteristic properties, rheometry, mooney viscosity, hardness, tensile, elongation, modulus, tear, low temperature properties, ozone resistance, aging properties, abrasion resistance, rebound resilience, set properties, electrical properties, tear resistance, dispersion, heat build up, swelling as per requirement /standards.	6	3	-	-
<i>Testing for finished products</i>	12	8	-	-
PC12. carry out testing for finished products as per requirement/guidelines provided.	4	4	-	-
PC13. carry out testing as per specific gravity, stress-strain properties (ts, m300, eb, tear), abrasion resistance (volume loss), set properties (compression, tension, permanent etc.), fatigue test (crack initiation, crack growth), rebound resilience, hardness testing (shore a/shore d), hysteresis properties.	4	2	-	-
PC14. carry out testing as per swelling properties, aging (hot air, humid, salt, oxygen bomb), low temperature properties, adhesion (fabric-rubber, rubber-rubber, rubber-metal), electrical properties, dimension and checks endurance, uniformity, x-ray analysis, in accordance is / iso / international standards and sop.	4	2	-	-
<i>Housekeeping and Safety</i>	12	10	-	-
PC15. ensure health & safety in laboratory	-	1	-	-
PC16. ensure availability of safety accessories including eye wash station.	-	1	-	-
PC17. ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.	-	2	-	-
PC18. handle the equipment properly	2	1	-	-
PC19. ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. conduct the experiments wearing the appropriate attire such as safety goggles, gloves, closed toe shoes, long pants, tied hair.	2	1	-	-
PC21. use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations	2	1	-	-
PC22. escalate matters in case of any accidents, spills etc.	2	1	-	-
PC23. comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).	2	-	-	-
NOS Total	70	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2303
NOS Name	Conduct testing of rubber products at various stages of production.
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	Lab Chemist
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/10/2026
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N2304: Analysis, Reporting and Recording of Test Results.

Description

This unit is about carrying out analysis, reporting and recording of test results.

Elements and Performance Criteria

Data analysis

To be competent, the user/individual on the job must be able to:

- PC1.** keep tools and accessories ready before starting the analysis
- PC2.** ensure that the accuracy and periodicity of the data captured in records is as per sop.
- PC3.** identify appropriate technique in evaluating result.
- PC4.** interpret the results correctly using the identified technique(s).

Recording and Reporting

To be competent, the user/individual on the job must be able to:

- PC5.** record and maintain data as per company standards (sop).
- PC6.** ensure that reports/records are accurate and clear.
- PC7.** release or hold the raw material as per finding for further processing.
- PC8.** take up the results of the findings with supplier/qc in-charge/appropriate authority.
- PC9.** ensure regular exchange of raw material testing, in process testing and end products testing to facilitate prompt action for establishing proves specifications, controlling processes and maintaining uniformity in the processes.
- PC10.** inform concerned persons for rectifications, if needed in specified time limit.

Housekeeping and Safety

To be competent, the user/individual on the job must be able to:

- PC11.** ensure health & safety in laboratory.
- PC12.** ensure availability of safety accessories including eye wash station.
- PC13.** ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.
- PC14.** handle the equipment properly.
- PC15.** ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.
- PC16.** conduct the experiments wearing the appropriate attire such as safety goggles, gloves, closed toe shoes, long pants, tied hair.
- PC17.** use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations.
- PC18.** escalate matters in case of any accidents, spills etc.
- PC19.** comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).

Material disposal

To be competent, the user/individual on the job must be able to:

- PC20.** carry out disposal of waste and left over tested material safely as per sop.

Qualification Pack

PC21. dispose all materials used in the experiment safely as per health and safety management system of the company.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys quality policies and acceptance standards for raw materials, processed and final product.
- KU2.** organisational coding system of raw material, compounds and products.
- KU3.** chemicals used in the industry and their function.
- KU4.** different quality management systems.
- KU5.** principles of good laboratory practices applicable in the workplace.
- KU6.** material disposal procedure, importance of appropriate disposal of material and implications of not following the material disposal procedure.
- KU7.** quality and damage checks to be done and importance of the same
- KU8.** importance of identifying non-conforming products and storage of the same.
- KU9.** risk and impact of not following defined procedures/work instructions.
- KU10.** escalation matrix for reporting identified issues.
- KU11.** types of documentation in organization and importance of the same.
- KU12.** records to be maintained and implications of non-maintenance of the same.
- KU13.** company manual and from where to attain it.
- KU14.** importance of housekeeping & good shop floor practices (e.g.3s/5s).
- KU15.** health, safety and environment guidelines, legislation and regulations as applicable.
- KU16.** personal protection(which protective equipment to be used and how).
- KU17.** impact of poor practices on health, safety and environment.
- KU18.** potential hazards and actions to minimize the same.
- KU19.** escalation matrix and escalation procedure for reporting hazards.
- KU20.** the usage of different fire extinguisher.
- KU21.** impact of various practices on cost, quality, productivity, delivery and safety.
- KU22.** handover/ takeover the equipment/ work area as per companys sop.
- KU23.** knowledge of basic chemistry and simple mathematical calculations.
- KU24.** knowledge on different standard reference material.
- KU25.** role of different raw materials in rubber compounding, processing/ product manufacturing and performance.
- KU26.** understanding of role of the testing equipments.
- KU27.** testing equipments and related test methods and purpose of tests
- KU28.** procedures for storing samples
- KU29.** specifications of materials tested and its importance in the release system
- KU30.** national/international standard test methods for different materials.
- KU31.** preparation of standard chemical reagents for testing.
- KU32.** standard method of drawing samples and preparing them for testing.

Qualification Pack

- KU33.** methods/techniques used for labeling samples.
- KU34.** the methods that can be used for controlling test variables.
- KU35.** implications (impact on internal/external customers) of defective products, materials or components.
- KU36.** the material safety data sheets (msds) for all the materials used for the experiments that one is conducting.
- KU37.** procedures for storing and retention period for samples.
- KU38.** factors that adversely affect integrity of the sample.
- KU39.** statistical analysis of test data.
- KU40.** how to obtain and interpret records, charts, specifications, equipment manuals, history/technical support reports and other documents.
- KU41.** methods and techniques involved in evaluating information.
- KU42.** use of computer/application software.
- KU43.** units of measurement.
- KU44.** response to emergencies e.g. power failures, fire and system failures and manual intervention to avoid disaster.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and communicate details of work done to appropriate people using written/typed report or computer based record/electronic mail
- GS2.** maintain proper records as per given format.
- GS3.** read and understand manuals, health and safety instructions, memos, reports, job cards etc.
- GS4.** read images, graphs, diagrams.
- GS5.** understand the various coding systems as per company norms.
- GS6.** communication with upstream and downstream teams.
- GS7.** communicate with job owners like sample originating section, supplier etc.
- GS8.** disclose information only to those who have the right and need to know it.
- GS9.** communicate confidential and sensitive information discretely to authorized person as per sop.
- GS10.** communicate with people in a form and manner and using language that is open and respectful.
- GS11.** resolve any difficulties in relationships with colleagues , or get help from an appropriate person, in a way that preserves goodwill and trust.
- GS12.** take responsibility for completing ones own work assignment.
- GS13.** take initiative to enhance/learn skills in oness area of work.
- GS14.** the capacity to learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse ones learning.
- GS15.** is open to new ways of doing things.
- GS16.** the capacity to envisage and articulate personal goals
- GS17.** avoid absenteeism .

Qualification Pack

- GS18.** act objectively , rather than impulsively or emotionally when faced with difficult/stressful or emotional situations.
- GS19.** be punctual.
- GS20.** take a decision for any change/issue based on earlier successes (documented previous history) on similar issues.
- GS21.** work out changes in case a new improved machine/equipment is added in the process or any new material /chemical is developed replacing existing one.
- GS22.** make changes in cycle time due to improved process.
- GS23.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management.
- GS24.** consult the peer group and superiors to arrive at a favourable decision.
- GS25.** use of standard available problem solving techniques for decision making.
- GS26.** review and analyze the process steps to check on system non adherence and non conformity.
- GS27.** review the current sop and other standards for continuous improvement to facilitate decision making.
- GS28.** take a calculated risk with minimum losses.
- GS29.** organize samples and records properly.
- GS30.** communicate results as per organizational procedure.
- GS31.** perform analysis in given time line.
- GS32.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required).
- GS33.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS34.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS35.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS36.** work towards fulfilling the customers requirement as per their demand.
- GS37.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level.
- GS38.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS39.** maintain good/cordial relation with customers.
- GS40.** work on the feedback received from customer regarding the product.
- GS41.** interpret quality for tested samples.
- GS42.** apply problem-solving approaches in different situations.
- GS43.** application of statistics.
- GS44.** use of computer/ application software.
- GS45.** proper collection of waste material.
- GS46.** identify defects in the material and communicate it at the earliest and suggest improvements(if any) in process/material based on experience.
- GS47.** identify any issues affecting the material, equipment or surroundings.
- GS48.** escalate issues that cannot be solved as per the troubleshooting/company manual

Qualification Pack

- GS49.** apply appropriate technique/method for various types of products to meet desired purpose
- GS50.** interpret data and analyse results.
- GS51.** suggest improvements(if any) in process/product/materials based on experience

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Data analysis</i>	24	3	-	-
PC1. keep tools and accessories ready before starting the analysis	6	3	-	-
PC2. ensure that the accuracy and periodicity of the data captured in records is as per sop.	6	-	-	-
PC3. identify appropriate technique in evaluating result.	6	-	-	-
PC4. interpret the results correctly using the identified technique(s).	6	-	-	-
<i>Recording and Reporting</i>	33	9	-	-
PC5. record and maintain data as per company standards (sop).	6	3	-	-
PC6. ensure that reports/records are accurate and clear.	5	3	-	-
PC7. release or hold the raw material as per finding for further processing.	5	3	-	-
PC8. take up the results of the findings with supplier/qc in-charge/appropriate authority.	6	-	-	-
PC9. ensure regular exchange of raw material testing, in process testing and end products testing to facilitate prompt action for establishing proves specifications, controlling processes and maintaining uniformity in the processes.	6	-	-	-
PC10. inform concerned persons for rectifications, if needed in specified time limit.	5	-	-	-
<i>Housekeeping and Safety</i>	9	12	-	-
PC11. ensure health & safety in laboratory.	-	2	-	-
PC12. ensure availability of safety accessories including eye wash station.	-	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. ensure availability of raw materials/laboratory reagents material safety data sheet (msds) in the laboratory.	-	2	-	-
PC14. handle the equipment properly.	1	1	-	-
PC15. ensure samples and chemicals are carefully handled to avoid accidental spillage of chemicals.	2	2	-	-
PC16. conduct the experiments wearing the appropriate attire such as safety goggles, gloves, closed toe shoes, long pants, tied hair.	1	1	-	-
PC17. use safety equipment such as fire extinguishers, fire blankets, and eye-wash stations.	2	1	-	-
PC18. escalate matters in case of any accidents, spills etc.	2	1	-	-
PC19. comply with health, safety, environment guidelines, regulations etc in accordance with international/national standards or organizational standards (sop).	1	-	-	-
<i>Material disposal</i>	4	6	-	-
PC20. carry out disposal of waste and left over tested material safely as per sop.	2	3	-	-
PC21. dispose all materials used in the experiment safely as per health and safety management system of the company.	2	3	-	-
NOS Total	70	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2304
NOS Name	Analysis, Reporting and Recording of Test Results.
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	Lab Chemist
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/10/2026
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

Qualification Pack

- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

Qualification Pack

- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

Qualification Pack

- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N2302.Collect rubber sample and prepare equipment	50	50	-	-	100	25
RSC/N2303.Conduct testing of rubber products at various stages of production.	70	30	-	-	100	30
RSC/N2304.Analysis, Reporting and Recording of Test Results.	70	30	-	-	100	25
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	240	210	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.